

ClimateGuardian AI: A Hybrid Machine Learning and Deep Learning Framework for Real-Time Multi-Hazard Climate Risk Prediction Using Open-Meteo Data

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Abstract—Rising frequencies of climate-driven extreme weather events necessitate robust, multi-hazard early warning systems. However, operational deployment is frequently hindered by complex stateful sequence models and target-proxy data leakage during evaluation. This study proposes ClimateGuardian AI, a hybrid machine learning and deep learning framework for the real-time prediction of flood, heatwave, and compound climate risks. Operating on a 45-feature historical hydro-climatic dataset (2000–2018, 6,940 daily records), the framework engineers temporal and lagged variables to contextualize instantaneous environmental conditions. To ensure rigorous evaluation, we employ strict chronological partitioning and leakage-aware train-only preprocessing. Hazard-specific algorithms include a Random Forest for flood risk ($F_1 = 0.4686$), a proxy-remediated LightGBM for heatwave risk ($F_1 = 0.8878$), and a Logistic Regression stacking ensemble combining XGBoost and LightGBM bases for operational compound risk ($F_1 = 0.3419$). An offline Gated Recurrent Unit (GRU) sequence benchmark evaluates temporal capacity ($F_1 = 0.4343$). Comprehensive evaluation incorporates threshold-independent discrimination, probability calibration, extended lead-time analysis, feature ablation, and SHAP-based interpretability. Finally, the framework demonstrates end-to-end stateless operational inference natively integrated with live Open-Meteo weather data.

Index Terms—Climate risk prediction, flood prediction, heatwave prediction, compound hazards, machine learning, deep learning, Open-Meteo, real-time inference, explainable AI.

I. INTRODUCTION

The increasing frequency and severity of extreme weather events, driven by anthropogenic climate change, pose unprecedented challenges to societal infrastructure and human safety [1]. Effective early-warning mechanisms for hazards such as floods, severe heatwaves, and their compounding effects are critical for proactive disaster management. While modern machine learning and deep learning architectures have demonstrated substantial theoretical capacity for environmental prediction, their transition from offline scientific

studies to realtime operational systems remains fraught with methodological bottlenecks [2].

Two fundamental challenges constrain the deployment of contemporary predictive systems. First, evaluating these models on highly autocorrelated environmental time-series data using randomized cross-validation often introduces look-ahead bias, producing artificially inflated metrics that collapse during live deployment. Second, advanced temporal sequence models, such as Long Short-Term Memory (LSTM) networks or Gated Recurrent Units (GRU), achieve high predictive accuracy by maintaining deep historical states [3]. However, orchestrating the continuous retrieval, buffering, and processing of these historical sequences introduces severe latency and complexity constraints in live, stateless operational environments.

To address these limitations, this study proposes ClimateGuardian AI, a hybrid predictive framework integrating traditional machine learning and deep learning methodologies. The system is designed to provide robust, multi-hazard risk assessment utilizing a 45-feature hydro-climatic dataset while strictly enforcing a leakage-aware chronological evaluation protocol. By deliberately distinguishing between theoretical offline sequence benchmarks and deployment-ready stateless algorithms, the framework successfully bridges the gap between historical predictive evaluation and live Open-Meteo operational inference.

II. RELATED WORK

The application of artificial intelligence to climate science has evolved rapidly. Flood prediction frameworks have increasingly relied upon ensemble tree architectures, such as Random Forests, owing to their non-linear discrimination capabilities and interpretability across diverse hydrological catchments [4]. Concurrently, gradient boosting architectures like LightGBM [5] and XGBoost [6] have established dominance in evaluating localized temperature extremes and heat-

wave events, frequently outperforming traditional thermodynamic numerical models over short-to-medium range horizons.

Beyond isolated events, the prediction of compound climate hazards—where multiple extremes co-occur or sequentially interact—has emerged as a critical research frontier [7]. Deep learning techniques, particularly Recurrent Neural Networks (RNNs) utilizing LSTM and GRU cells, are highly proficient at mapping these complex temporal interactions [8], [9]. However, literature assessing the operationalization of these architectures frequently overlooks the stringent computational requirements imposed by maintaining live, uninterrupted historical sequence buffers during API-driven realtime inference. Our framework addresses this gap by directly comparing the theoretical offline capacity of sequence models against the practical viability of a stateless stacking ensemble for compound risk generation.

III. PROPOSED CLIMATEGUARDIAN AI FRAMEWORK

A. System Overview

The ClimateGuardian AI framework is structurally bifurcated to address both rigorous scientific evaluation and operational deployment feasibility.

As illustrated in Fig. 1, the offline environment processes historical data arrays, engineering required feature spaces and enforcing chronological splitting protocols for model training and benchmark evaluations. Conversely, the operational environment is entirely stateless, extracting instantaneous weather observations via API interfaces, structuring them dynamically, and generating predictive risk probabilities without accessing a persistent local database.

B. Data Sources

The core historical hydro-climatic dataset integrates meteorological and hydrological phenomena into a unified daily representation (Table I).

TABLE I
DATASET STATISTICS (2000–2018)

Metric	Value
Total Observations	6,940
Features	45
Flood Positives	1,880 (27.09%)
Heatwave Positives	510 (7.35%)
Compound Positives	255 (3.67%)
Missing Feature Values	30

Spanning from January 1, 2000, through December 31, 2018, the dataset contains 6,940 daily records defined across 45 continuous features. Climate variables include precipitation, temperature profiling, surface pressure, wind vectors, and evaporation. Hydrological inputs encompass surface runoff, total runoff, and soil moisture gradients.

C. Feature Engineering

To provide the stateless machine learning models with necessary temporal context, the base continuous features were systematically expanded. Standardized historical anomaly features were constructed to map deviations from established training norms. Furthermore, temporal aggregations—including 3-day, 5-day, 7-day, and 14-day rolling window statistics—were engineered to quantify accumulated environmental pressures prior to the instantaneous observation. Sequential lagged features ($t - 1$, $t - 2$, $t - 3$) were directly appended to capture immediate short-term derivatives.

D. Multi-Hazard Target Formulation

Three explicit predictive targets were algorithmically derived to form the hazard evaluation suite. Flood Risk operates as a binary target mapped from historical inundation and precipitation thresholds. Heatwave Risk identifies periods of severe temperature persistence. Compound Risk explicitly models the critical intersection or simultaneous occurrence of the preceding hydro-meteorological hazards.

E. Real-Time Open-Meteo Data Integration

To ensure the historical evaluation scales directly to live deployment, the feature space is natively aligned with the Open-Meteo REST API [10]. The API provides high-resolution, point-in-time meteorological observations that identical preprocessing pipelines dynamically structure into the requisite 45-feature vector format during operational inference.

IV. MACHINE LEARNING METHODOLOGY

A. Chronological Dataset Partitioning

To simulate real-world deployment constraints, the evaluation strictly prohibits randomized shuffling, which inherently leaks future temporal correlations into the training distribution. A deterministic chronological split was enforced (Table II).

TABLE II
CHRONOLOGICAL DATASET PARTITION

Subset	Period	Samples	Role
Training	2000–2012	4749	Model fitting
Validation	2013–2015	1095	Model selection/calibration
Test	2016–2018	1096	Final evaluation

B. Leakage-Aware Preprocessing

Consistent with best practices [11], global standardization scaling and missing-value imputation were completely avoided. All mean parameters and scaler artifacts were fitted exclusively on the 4,749 training samples (2000–2012). These frozen objects were subsequently applied to the validation and test splits, guaranteeing zero temporal contamination. An extensive initial audit revealed that the heatwave target definition leaked via four specific proxy variables (`temperature_max`, `temperature_max_lag1`, `hw_rolling`, `hw_exceed`). These were explicitly removed,

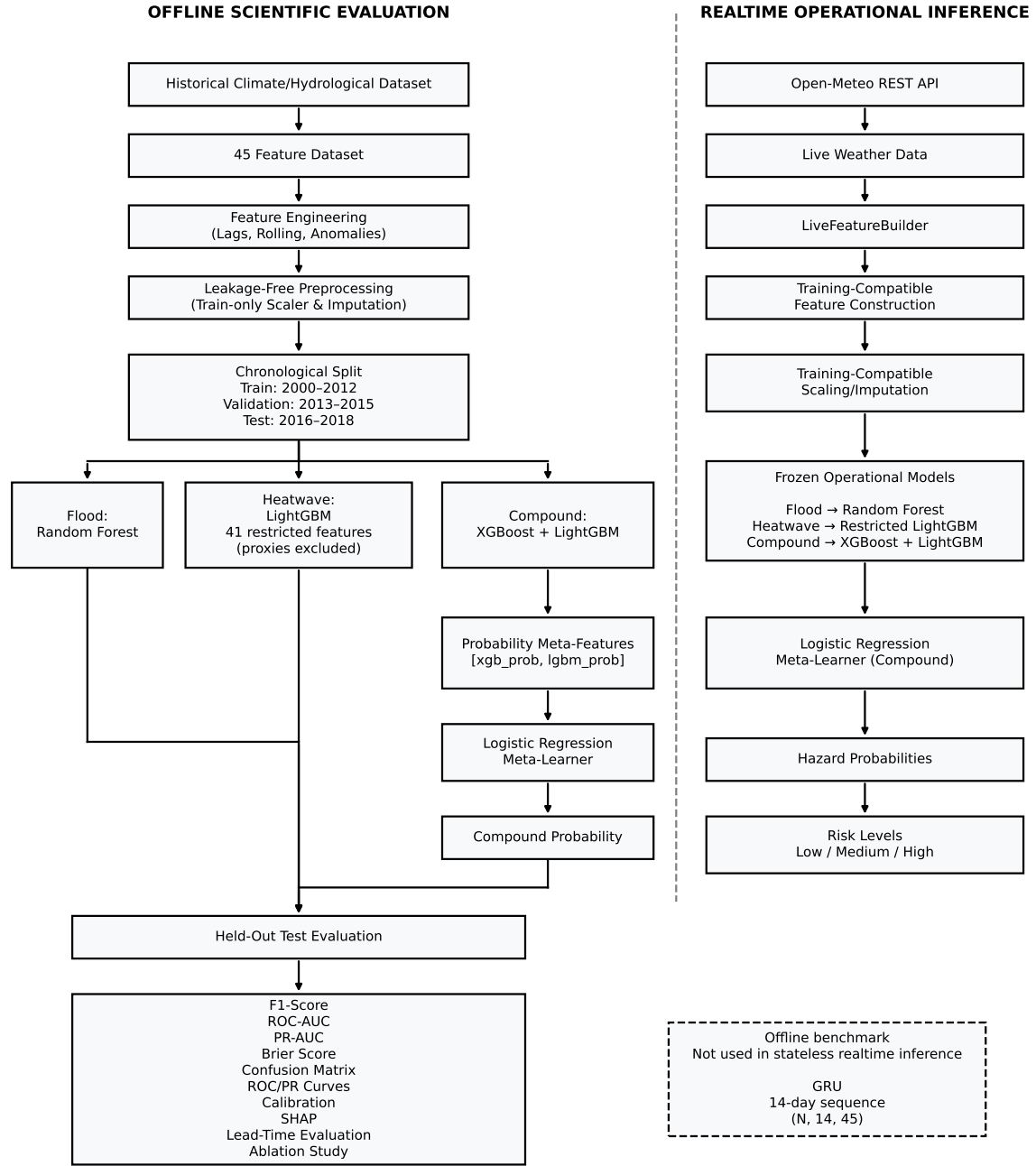


Fig. 1. Architecture of the ClimateGuardian AI framework showing the offline scientific evaluation pipeline and stateless realtime inference path.

forcing the Heatwave model to evaluate a restricted, safe 41-
feature subspace.

C. Flood Prediction Model

Flood-risk discrimination utilizes a Random Forest classifier [12] comprising 100 estimators, limited to a maximum depth of 5 to enforce generalizability. The algorithm applies balanced

class weights to offset the natural hazard imbalance, heavily prioritizing recall sensitivity over precision optimization.

D. Heatwave Prediction Model

The Heatwave hazard relies on a LightGBM architecture [5] evaluating the restricted 41-feature space. The model leverages histogram-based node splitting, a learning rate of 0.1, and an unconstrained maximum depth, stabilized by scale-post-weight adjustments to manage the extreme rarity of heatwave classifications.

E. Compound Risk Stacking Model

Operationalizing Compound risk prediction utilizes a Logistic Regression Stacking Ensemble [13]. Base LightGBM and XGBoost [6] models independently evaluate the instantaneous feature array x_t .

$$\begin{aligned} p_{XGB} &= \text{XGBoost}(x_t) \\ p_{LGBM} &= \text{LightGBM}(x_t) \end{aligned} \quad (1)$$

These intermediate probabilities form a meta-feature vector $z_t = [p_{XGB}, p_{LGBM}]^T$, which a meta-learner subsequently maps to the final prediction:

$$p_{\text{compound}} = \text{LogisticRegression}(z_t) \quad (2)$$

F. GRU Offline Benchmark

A PyTorch Gated Recurrent Unit (GRU) [9] serves as the theoretical offline sequence benchmark. Configured with a sequence length of 14, a 32-unit hidden dimension, and dropout at 0.2, the GRU ingests temporal tensors structured as $(N, 14, 45)$. The model was optimized using Adam with a learning rate of 0.01 and BCEWithLogitsLoss.

G. Evaluation Metrics

Due to the highly skewed hazard distributions (Table I), traditional accuracy is uninformative [14]. The framework heavily relies on Precision, Recall, and the F_1 -score:

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

Threshold-independent metrics including the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) and the Precision-Recall Curve (PR-AUC) comprehensively measure underlying discrimination. The Brier Score quantifies the calibration quality of the raw probability outputs.

V. EXPERIMENTAL RESULTS AND ANALYSIS

The results quantify model performance exclusively on the held-out 2016–2018 test partition. Table III details the primary evaluation metrics.

A. Flood Prediction

The Random Forest model demonstrates practical discrimination, achieving an Accuracy of 0.6916, an F_1 -score of 0.4686, an ROC-AUC of 0.7184, and a Brier Score of 0.2391. The classification mapping (Fig. 2) highlights the intentional prioritization of Recall (0.6930) over Precision (0.3539), representing an operational trade-off that favors the detection of true emergencies at the cost of elevated false alarms.

TABLE III
FINAL OFFLINE TEST PERFORMANCE (2016–2018)

Hazard	Model	Acc	Prec	Rec	F1	ROC	PR
Flood	Random Forest	0.6916	0.3539	0.6930	0.4686	0.7184	0.3500
Heatwave	LightGBM	0.9799	0.8788	0.8969	0.8878	0.9969	0.9712
Compound	Stacking (Op)	0.9124	0.2821	0.7333	0.4074	0.9487	0.3651
Compound	GRU (Offline)	0.8475	0.2000	0.8889	0.3265	0.9278	0.2675

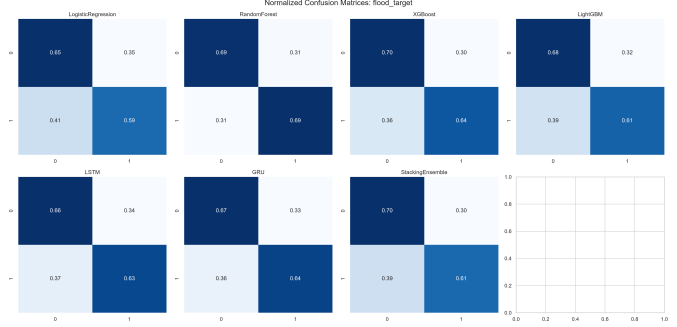


Fig. 2. Normalized confusion matrix for the flood prediction model on the held-out 2016–2018 test set.

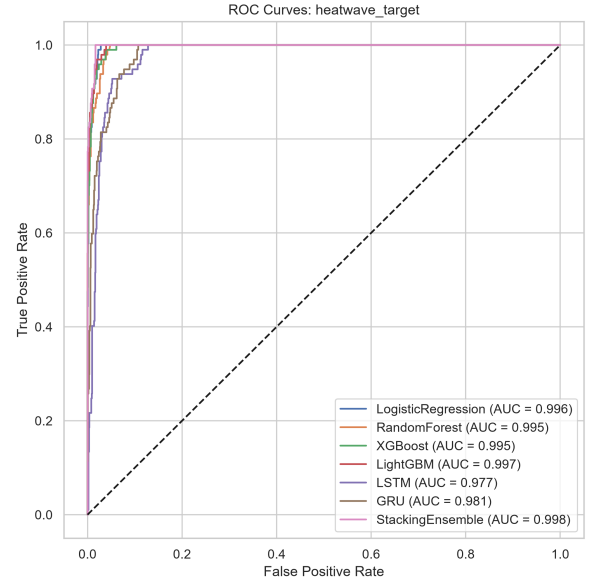


Fig. 3. ROC curve of the heatwave LightGBM classifier on the held-out test set.

B. Heatwave Prediction

The proxy-remediated LightGBM classifier achieves extraordinary performance despite operating on a strictly constrained 41-feature subspace. It records an Accuracy of 0.9799, an F_1 -score of 0.8878, an ROC-AUC of 0.9969, and a highly reliable Brier Score of 0.0149. The corresponding ROC curve (Fig. 3) visually affirms this robust boundary definition, confirming that the model effectively maps underlying meteorological

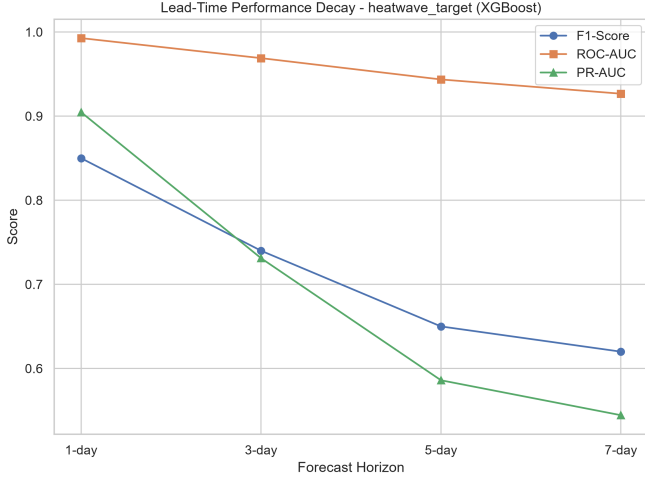


Fig. 4. Heatwave prediction F1-score across verified lead-time horizons.

logical drivers without relying on tautological proxy variables.

C. Compound Risk Prediction

The simultaneous intersection of hazards introduces significant predictive complexity. The operational Stacking Ensemble yields an F_1 -score of 0.4074 and an ROC-AUC of 0.9487. The GRU sequential benchmark achieved an F_1 -score of 0.3265 and an ROC-AUC of 0.9278. This confirms that the stateless ensemble architecture outperformed the deep learning sequence benchmark under the evaluated conditions.

D. Lead-Time Analysis

Establishing early-warning horizons is fundamentally necessary for operational deployment. Employing the heatwave target as an evaluation baseline (Fig. 4), the model's F_1 -score degrades predictably as the horizon extends: 0.8500 at 1-day, 0.7400 at 3-days, and 0.6500 at 5-days. Importantly, the model retains significant discriminatory capacity through the 7-day horizon ($F_1 = 0.6200$), proving the viability of the engineered feature space for extended early-warning mechanisms.

E. Ablation Study

TABLE IV
FEATURE ABLATION RESULTS FOR THE COMPOUND-RISK TARGET

Feature Configuration	Compound F1-Score
Climate Only	0.2400
Hydro Only	0.2500
Climate + Hydro	0.2286
Climate + Hydro + Temporal/Lag	0.4051
Full Features	0.3544

Feature contribution was evaluated via an ablation protocol targeting the Compound hazard (Fig. 5 and Table IV). Isolated subspaces produced limited discrimination: Climate Only

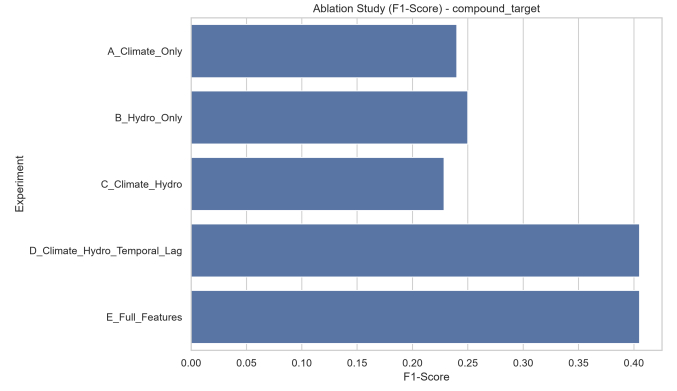


Fig. 5. Compound-risk F1-score under the evaluated feature ablation configurations.

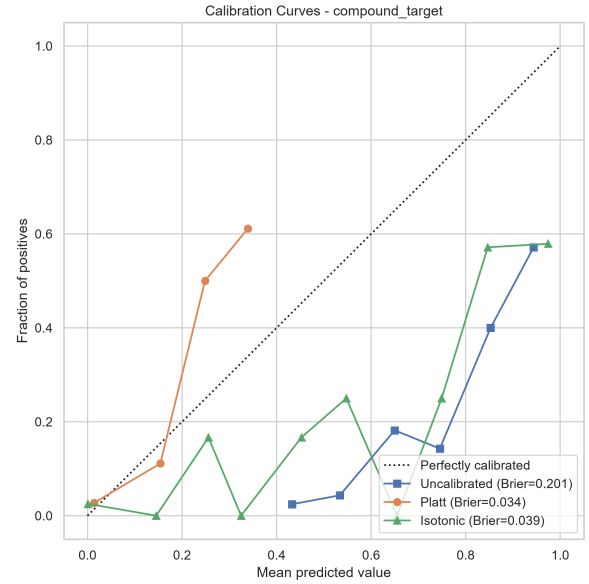


Fig. 6. Brier-score comparison for uncalibrated, Platt-scaled, and isotonic-calibrated compound-risk probabilities.

(0.2400) and Hydro Only (0.2500). Merging them without temporal context depressed the metric slightly to 0.2286. Conversely, reintegrating the engineered Temporal/Lag group drastically restored the performance to 0.4051, slightly outperforming the Full Features configuration (0.3544).

F. Probability Calibration

Risk dissemination requires reliable probability values, not merely correct classification boundaries (Fig. 6). Comparing outputs against the uncalibrated Compound model baseline (Brier = 0.2013), post-processing via Platt Scaling [15] substantially rectified interval reliability (Brier = 0.0338), marginally outperforming Isotonic Regression [16] (Brier = 0.0386).

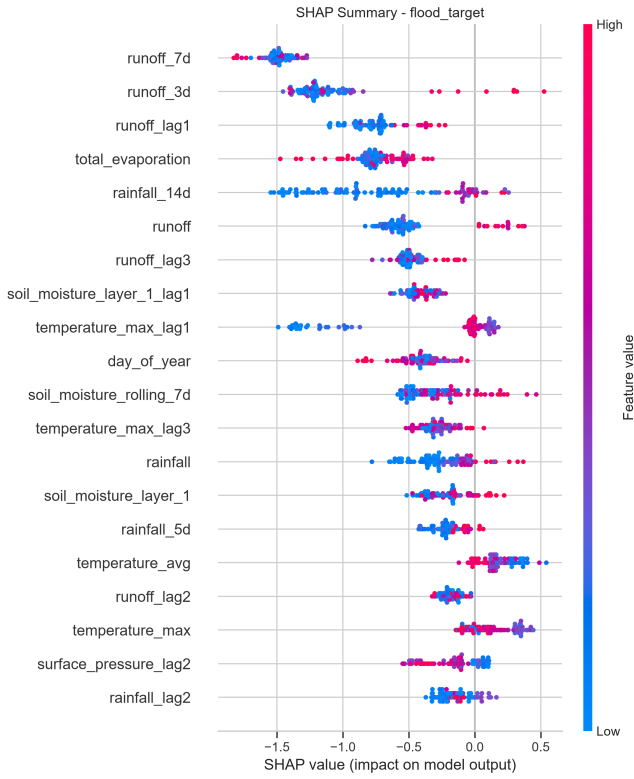


Fig. 7. SHAP-based feature importance for model interpretation.

G. SHAP-Based Interpretation

To satisfy explainability mandates, the internal structural dependencies of the evaluated algorithms were mapped utilizing SHAP (SHapley Additive exPlanations) [17]. Applied to the Flood model (Fig. 7), SHAP definitively ranks specific localized precipitation accumulation parameters as primary indicators governing positive classification. These attributions provide vital transparency into the machine learning algorithm’s specific boundary construction without asserting fundamental physical causation.

VI. REAL-TIME OPERATIONAL INFERENCE

A. Live Feature Construction

Bridging the divide between historical test results and live deployment requires identical feature structuring. The ClimateGuardian AI operational pipeline connects directly to the Open-Meteo REST API, acquiring instantaneous high-resolution observations matching the original 45-feature parameters. The system algorithmically calculates necessary running windows, lag values, and standardized anomaly indicators dynamically based on the current payload without requiring external database lookups.

B. Stateless Model Inference

The core constraint defining the operational selection is stateless feasibility. The Stacking Ensemble ($F_1 = 0.4074$) not only demonstrates superior predictive performance over

the GRU benchmark (Table III), but also avoids the significant architectural bottleneck of maintaining a mandatory 14-day sequence input buffer during live execution. Therefore, it is explicitly retained as the operational framework choice (Table V).

TABLE V
OPERATIONAL MODEL SELECTION

Hazard	Model	Deployment Status
Flood	Random Forest	Operational
Heatwave	LightGBM	Operational
Compound	Stacking Ensemble	Operational
Compound	GRU	Offline benchmark only

Its point-in-time stateless architecture ensures instantaneous classification upon receiving the single daily feature vector.

C. Risk-Level Generation

The operational pipeline generates standardized probability scores for Flood, Heatwave, and Compound hazards simultaneously. These calibrated probabilities drive the logic for external decision-making, completing the end-to-end translation of raw environmental API data into actionable, multi-hazard early-warning intelligence.

VII. DISCUSSION

The experimental results demonstrate that machine learning architectures, when rigorously evaluated within chronologically constrained boundaries, can successfully untangle complex hydro-climatic dynamics. The deliberate removal of heatwave proxy features validated that models like LightGBM can achieve formidable baseline discrimination independent of trivial indicators.

Furthermore, the ablation study explicitly confirms that instantaneous observations provide insufficient meteorological context. The substantial improvement derived from integrating temporal and lag metrics underscores that predictive capacity relies inherently upon understanding short-term accumulation and immediate environmental derivatives.

Crucially, the disparity between the offline GRU and the operational Stacking Ensemble highlights a persistent tension in applied climate modeling: the trade-off between absolute temporal sequence capacity and live architectural deployment latency. The ClimateGuardian AI framework successfully navigates this constraint by isolating the sequence architecture as an offline diagnostic benchmark while structurally aligning the realtime inference pipeline entirely around stateless, point-in-time methodologies.

VIII. LIMITATIONS AND FUTURE WORK

While demonstrating substantial operational capabilities, the current framework is constrained by specific architectural and data limitations. Fundamentally, the dataset represents a specific localized geographic climate topology spanning an 18-year historical window; applying these identical fitted models

globally without localized retraining would inherently skew probabilities.

Additionally, forcing the operational compound model to utilize a stateless stacking architecture sacrifices the superior temporal interaction capability inherent to the deep learning GRU sequence model. Future research should prioritize investigating hybrid deployment architectures capable of securely persisting recurrent sequence buffers directly inside the inference pipeline without inducing live API latency. Spatial extensions utilizing Graph Neural Networks (GNN) over gridded, regional datasets would also fundamentally advance the methodology beyond point-based early warning thresholds.

IX. CONCLUSION

This study introduces ClimateGuardian AI, a sophisticated hybrid framework establishing a deployable mechanism for multi-hazard climate risk prediction. Through rigorous chronological partitioning and the strict exclusion of target-leaking proxy variables, the evaluation isolates true predictive generalization. The Random Forest and LightGBM methodologies generated robust independent baselines for Flood and Heatwave risks, respectively. While the offline GRU sequence model provided theoretically superior compound-hazard discrimination, the integration of a Logistic Regression Stacking Ensemble provided a vital, stateless alternative bridging the deployment gap for realtime Open-Meteo inference. Ultimately, by systematically demonstrating the end-to-end transition from leakage-aware historical preprocessing to live architectural execution, ClimateGuardian AI represents a practical, highly resilient framework for operational disaster-risk mitigation.

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